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VOLUME Th2

Th2. Self-adjointness

Th1–Th6: hermiticity, self-adjointness, certificates, the spectral Li formula, block reduction, the Guinand–Weil trace formula

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Th2. Finite-dimensional self-adjointness

Theorem Th2 (V8).

A symmetric operator on $\mathbb{C}^N$ is essentially self-adjoint: defect indices (0, 0); e^{-iHt} is unitary; $\mathrm{Im}\langle\psi, H\psi\rangle = 0$.

> ***Proof.*** *On $\mathbb{C}^N$ every bounded symmetric operator is self-adjoint: $D(H) = \mathbb{C}^N$ and the defect spaces $\ker(H^* - iI)$, $\ker(H^* + iI)$ are empty, since $\|(H \mp iI)\psi\|^2 = \|H\psi\|^2 + \|\psi\|^2 \geq \|\psi\|^2 > 0$ (the cross-term vanishes by symmetry). Consequences: real spectrum, e^{-iHt} a unitary group, $\langle\psi, H\psi\rangle$ real. Numerically: the minimal singular value of $H - iI$ is 3.9 (theory: $\sqrt{1 + \min\|H\psi\|^2} \geq 1$), the unitarity drift of e^{-iHt} is 2.2e-16 over 1000 steps. The limit part ($L \to \infty$, essential self-adjointness of an unbounded operator) is the open obligation O03.*

Th2.1 Certificate and boundaries

Certificate: test_th2_selfadjoint.py checks three equivalent manifestations of self-adjointness — empty defects, unitary dynamics, real form — on the protocol gate; all three hold at machine floors. Practical value: the unitarity of e^{-iHt} underpins the dynamics campaigns (V17): the energy conservation 1.8e-15 in V17 is a direct corollary of Th2. The gap "finite-dim proved / limit open" is recorded honestly: in the infinite-dimensional problem symmetry is not enough, boundary conditions are needed (open boundaries are natural, but the strict convergence of resonances is an open problem of the program).

Boundaries: the theorem is finite-dimensional; the transfer to $L \to \infty$ requires estimates of the essential spectrum and resonance geometry — see O03 in the obligation DAG.